Project Design Phase-I
Solution Architecture

Date	8 March 2026
Team ID	699ee17ef292113d819d8e62
Project Name	Predictive Pulse: Harnessing Machine Learning for Blood Pressure Analysis
Maximum Marks	4 Marks

Solution Architecture:

- The solution architecture of the Predictive Pulse system is a multi-layered framework designed to bridge the gap between raw medical data and actionable clinical intelligence.

- It serves as a comprehensive process that describes the structure, behaviour, and characteristics of the hypertension prediction tool to healthcare stakeholders. The architecture is meticulously organized into eight distinct layers, ranging from initial data acquisition to final deployment on secure medical servers.

- By integrating preprocessing, machine learning model persistence, and a Flask-based web interface, the architecture ensures that the transition from a patient's clinical parameters to a life-saving diagnosis is seamless and computationally efficient.

- This systematic design supports healthcare professionals by providing a reliable decision-making tool that can identify hypertension stages—ranging from Normal to Hypertensive Crisis—instantly.

ARCHITECTURAL COMPONENTS:

1.Data Collection Layer:

- The foundation of the Predictive Pulse system lies in its ability to handle diverse medical datasets. This layer is responsible for gathering multi-dimensional parameters such as systolic and diastolic blood pressure readings, existing symptoms, current medication status, and detailed patient history.

2 Data Preparation Layer:

- Raw medical data is frequently incomplete or noisy, necessitating a rigorous preparation layer. This layer employs cleaning techniques to handle missing values that could skew the model's accuracy.

- Key steps include categorical data encoding where qualitative symptoms are converted into numerical format and data normalization to ensure that variables with larger ranges do not dominate the learning process.

- Furthermore, feature selection is utilized to identify only the most relevant parameters, reducing computational overhead and preventing the model from learning irrelevant patterns.

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3 Data Splitting and Machine Learning Layer:

- To ensure the system's reliability, the dataset is divided into training and testing subsets. The training data serves as the educational base for the algorithms, while the testing data provides an unbiased evaluation of the model's performance on unseen patients.

- The machine learning layer implements multiple supervised algorithms to predict risk. These models are rigorously compared, and the highest-performing algorithm determined by metrics like precision and recall is saved using model persistence techniques for real-time application usage.

4 Prediction Engine and Web Application Layer:

- The prediction engine is the "brain" of the system, where the pre-trained model processes live inputs to generate a hypertension risk classification. It categorizes patients into four specific clinical stages: Normal, Stage-1, Stage-2, or Hypertensive Crisis.

- This engine is accessed through a Flask-based web application, which serves as the primary user interface for healthcare providers. Providers can input parameters directly through this interface, and the system communicates with the backend prediction engine to display diagnostic results instantly

Solution Architecture Diagram: